

# Error Analysis for Solving a Linear System by Cholesky Factorization

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## Abstract

Following Demmel [Dem81] with some lemmas improved by Rump and Jeannerod [RJ14] and then implemented in libValidSDP by Roux [Rou16], we derive the backward error analysis for solving a s.p.d. linear system  $Ax = b$  with fully accurate consideration of underflows. We do this as a guide to what will be mechanized in Rocq, or what will be documentation of what will have been mechanized.

## 1 Preliminaries

*Almost all of this section is quoted directly from Demmel (1981).*

Let  $\epsilon$  be the rounding error of the arithmetic, and  $\lambda$  be the underflow threshold. Suppose, for example, a binary floating point number is represented as  $f \cdot 2^e$  where  $f$  lies between 1 and 2 with an  $n$  bit fraction, and  $e$  is an integer exponent in the range  $e_{\min} \leq e \leq e_{\max}$ . Then  $\epsilon = 2^{-n}$  (the difference between 1 and the next largest number). The underflow threshold is  $\lambda = 2^{e_{\min}}$ ; this is the smallest ... normalized number for G.U. ... Denormalized numbers lie between 0 and  $2^{e_{\min}}$  in G.U. arithmetic for which the smallest nonzero number is thus  $\mu = \epsilon\lambda$ . We will give a more precise model of rounding and underflow errors in the next section.

Let  $\bullet$  be one of the operations  $(+, -, *, /)$  and let  $fl(a \bullet b)$  denote the floating point result of the indicated computation. ... To take underflow into account, we write ([KP79]),

$$fl(a \bullet b) = (a \bullet b)(1 + e) + \eta \quad \text{unless } a \bullet b \text{ overflows.} \quad (11)$$

In the case of G.U. we have the following constraints on  $e$  and  $\eta$ :

- (1)  $|e| \leq \epsilon$  and  $|\eta| \leq \lambda\epsilon$ ,
- (2)  $\eta \times e = 0$
- (3)  $\eta = 0$  if  $\bullet$  is either addition or subtraction.

We define the function  $UN(x)$  to be

$$UN(x) = \begin{cases} 1 & \text{if } x \text{ underflows } (x < \lambda) \\ 0 & \text{if it does not} \end{cases}$$

$\|A\|$  and  $\|b\|$  denote the norms of matrix  $A$  and vector  $b$ .  $\|A\|_{\infty}$  denotes the infinity norm of  $A$ , and similarly for  $\|b\|_{\infty}$ .  $k(A) = \|A\|_{\infty}\|A^{-1}\|_{\infty}$  denotes the condition number of the matrix  $A$ .

$|A|$  ( $|b|$ ) denotes the matrix (vector) whose entries are the absolute values of the corresponding entries in  $A$  ( $b$ ). Inequalities like  $|A| > |B|$  ( $|a| > |b|$ ) are meant componentwise.

**Approach.** We use backward error analysis. Thus, when trying to solve

$$Ax = b \quad (12)$$

for  $x$  we obtain an approximation  $\hat{x} = x + \delta x$  which satisfies the perturbed problem

$$(A + \delta A)\hat{x} = b + \delta b. \quad (13)$$

The task of backwards error analysis is to obtain bounds on  $\delta A$  and  $\delta b$ . These bounds can be used in turn to bound the residual  $r$ :

$$r = A\hat{x} - b = -\delta A\hat{x} + \delta b \quad (14)$$

## 2 Comparison with Demmel (1981)

**Demmel's Proposition 2** is as follows. As we explain in the next sections, we'll use something slightly different.

To analyze solving a triangular system, we need another expression for the error in computing an inner product:

$$\text{float}(\sum_{i=1}^n a_i b_i) = \sum_{i=1}^n a_i b_i (1 + E_i) + \eta . \quad (\text{demmel.31})$$

In the absence of underflow we have

$$\begin{aligned} |E_1| &\leq n\epsilon \\ |E_j| &\leq (n + 2 - j)\epsilon \quad \text{if } j > 1 . \end{aligned} \quad (\text{demmel.32})$$

In the case of G.U.<sup>1</sup> we have the same bounds on the  $|E_i|$ , and

$$|\eta| \leq n\lambda\epsilon \quad (\text{demmel.34})$$

**Demmel's Lemma 4: Error Analysis of Solving a Triangular System.** *As we explain in the next sections, we will use a variation of this lemma.*

Consider the lower triangular system  $Ly = b$  where  $L$  does not necessarily have ones on the diagonal. (The analyses when only ones are on the diagonal or  $L$  is upper triangular are similar.) Then in solving  $Ly = b$  using the program in Appendix 2 (where inner products are accumulated with roundoff error  $\epsilon_s$  and smallest number  $\mu_s$  we obtain  $\hat{y} = y + \delta y$  which actually satisfies  $(L + \delta L)\hat{y} = b + \delta b$ , where

$$|\delta L_{ij}| \leq |L_{ij}| \cdot \begin{cases} \epsilon & \text{if } i = j = 1 \\ (i - 1)\epsilon_s & \text{if } i > j = 1 \\ (i + 1 - j)\epsilon_s & \text{if } i > j > 1 \\ \epsilon + \epsilon_s & \text{if } i = j > 1 \end{cases} \quad (\text{demmel.80})$$

independent of underflow.

In the absence of underflow

$$\delta b = 0 . \quad (\text{demmel.81})$$

Using G.U.

$$\begin{aligned} \text{if } i = 1 \quad |\delta b_i| &\leq \lambda\epsilon |L_{11}| UN(y_1) && \text{if } y_1 \text{ underflows} \\ \text{if } i > 1 \quad |\delta b_i| &\leq i \cdot \lambda_s \epsilon_s + \lambda\epsilon |L_{ii}| UN(y_i) && \begin{array}{l} \text{intermediate underflows} \\ \text{if } y_i \text{ underflows} \end{array} \end{aligned} \quad (\text{demmel.82})$$

If  $L_{ii}$  is known to be 1, the only changes in the above bounds (besides substituting 1 for  $L_{ii}$ ) are that  $E_{11} = \delta b_1 = 0$  independent of underflow.

## 3 Error analysis of Cholesky Solve

Instead of Demmel's Proposition 2, we will be using the following lemma:

**Rump/Jeannerod Lemma 2.1** (LibValidSDP lemma\_2\_1): Analysis of  $(c - \sum a_i b_i)/b_k$ . Let  $\hat{y} = fl((c - \sum_{0 \leq i < k} a_i b_i)/b_k)$  computed by repeated subtraction left to right. Assume  $b_k \neq 0$ . Then

$$|b_k \hat{y} - (c - \sum_{0 \leq i < k} a_i b_i)| < (k + 1)\epsilon |b_k \hat{y}| \sum_{0 \leq i < k} |a_i b_i| + (1 + (k + 1)\epsilon)(k + 1 + |b_k|)\lambda\epsilon. \quad (\text{lemma2.1})$$

Demmel's Lemma 4 provides fine-tuned bounds for four different cases, depending on whether  $i > j$  and whether  $j > 0$ . In all these cases we will prove a weaker bound because that allows us to re-use the Rump/Jeannerod Lemma 2.1 for  $c - \sum x_i y_i$ .

<sup>1</sup>Editor's note: G.U. = Gradual Underflow, that is, a floating-point format that supports subnormal numbers. The technical report also covers the case of "S.Z." or "store zero", which we omit here in all lemmas and theorems.

**Theorem Demmel\_L4\_adapted:**

$$\begin{aligned}
|\delta L_{ij}| &\leq |L_{ij}|n\epsilon \quad \text{and} \quad (\text{demmel\_lemma\_4}) \\
\text{if } i = 0 \quad |\delta b_i| &\leq \lambda\epsilon|L_{00}|UN(y_0) \quad \text{if } y_0 \text{ underflows} \\
\text{if } i > 0 \quad |\delta b_i| &\leq (1 + (i + 1)\epsilon)(i + 1) \cdot \lambda_s\epsilon_s + \text{intermediate underflows} \quad . \quad (1) \\
&\quad (1 + (i + 1)\epsilon)\lambda\epsilon|L_{ii}|UN(y_0) \quad \text{if } y_i \text{ underflows}
\end{aligned}$$

The factor  $(1 + (i + 1)\epsilon)$  was omitted from Demmel's original but is required for correctness.

**Proof.** From libValidSDP lemma\_2\_1\_aux, by a derivation of several lines. Q.E.D.

## 4 Solving a linear system by Cholesky+Substitution

We now derive mixed error bounds for the floating-point result  $\hat{x}$  of solving  $Ax = b$  by Cholesky decomposition  $A = R^T R$  followed by forward substitution  $R^T y = b$  then back substitution  $Rx = y$ .

**Lemma th\_2\_3\_aux3: Accuracy of Cholesky decomposition.** Suppose  $A$  is an  $n \times n$  symmetric positive definite floating-point matrix, with  $(n + 1)\epsilon < 1$ . Let  $\hat{R}$  be the result of floating-point Cholesky decomposition (inner-product method) of  $A$  into  $\hat{R}^T \hat{R}$ . Let  $\delta A = A - \hat{R}^T \hat{R}$ . Let the vector  $R_j$  be the  $j$ th column of  $R$ . Let  $a_{\max}$  be the maximum diagonal element of  $|A|$ . Then

$$|\delta A_{ij}| < (\min(i, j) + 2)\epsilon \|R_i\|_2 \|R_j\|_2 + 4\lambda\epsilon(n + 1 + a_{\max}). \quad (2.3aux3)$$

**Proof.** See Rump+Jeannerod, or see libValidSDP. Q.E.D.

**Remark.** Let  $A$  and  $\hat{R}$  be as above. Then

$$\|\hat{R}_j\|_2 \leq \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1 - (j + 2)\epsilon)}. \quad (2.3aux1)$$

**Proof.** LibValidSDP th\_2\_3\_aux1.

**Corollary.** Combining (2.3aux1) and (2.3aux2), we have,

$$\begin{aligned}
|\delta A_{ij}| &< (\min(i, j) + 2)\epsilon \sqrt{(A_{ii} + 2i\lambda\epsilon)/(1 - (i + 2)\epsilon)} \\
&\quad \cdot \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1 - (j + 2)\epsilon)} + 4\lambda\epsilon(n + 1 + a_{\max}) \\
&\leq (n + 1)\epsilon \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} \\
&\quad \cdot \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} + 4\lambda\epsilon(n + 1 + a_{\max}) \\
&= (n + 1)\epsilon(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon) + 4\lambda\epsilon(n + 1 + a_{\max})
\end{aligned} \quad (\text{ch\_back\_err})$$

**Lemma.** We obtain bounds on  $\delta R^T \hat{R}$  by combining (2.3aux1) and (demmel\_lemma\_4):

$$\begin{aligned}
|(\delta R^T \hat{R})_{ij}| &\leq (|\delta R|^T |\hat{R}|)_{ij} \\
&\leq \sum_k (k + 1)\epsilon |\hat{R}_{ki}| \cdot |\hat{R}_{kj}| \quad (\text{by Lemma 0.4}) \\
&\leq n\epsilon \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 \quad (\text{by Cauchy - Schwartz}) \\
&\leq n\epsilon \sqrt{(A_{ii} + 2i\lambda\epsilon)/(1 - (i + 2)\epsilon)} \sqrt{(A_{jj} + 2j\lambda\epsilon)/(1 - (j + 2)\epsilon)} \quad (\text{by 2.3aux1}) \\
&\leq n\epsilon \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} \sqrt{(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)} \\
&= n\epsilon(a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon)
\end{aligned} \quad (\text{csolve\_back\_err\_aux1})$$

The same bound is obtained on  $\hat{R}\delta R$ .

**Lemma.** Bounds on  $\delta R^T \delta R$  as follows:

$$\begin{aligned}
|(\delta R^T \delta R)_{ij}| &\leq (|\delta R^T| \cdot |\delta R|)_{ij} \\
&\leq \sum_k |\delta R^T|_{ik} |\delta R|_{kj} \\
&= \sum_k |\delta R|_{ki} |\delta R|_{kj} \\
&\leq \sum_k (k + 1)^2 \epsilon^2 |\hat{R}_{ki}| \cdot |\hat{R}_{kj}| \\
&\leq n^2 \epsilon^2 \|\hat{R}_i\|_2 \|\hat{R}_j\|_2 \\
&\leq n^2 \epsilon^2 (a_{\max} + 2(n - 1)\lambda\epsilon)/(1 - (n + 1)\epsilon).
\end{aligned} \quad (\text{csolve\_back\_err\_aux2})$$

**Theorem: Cholesky factor + substitution.** Suppose  $A$  is an  $n \times n$  symmetric positive definite floating-point matrix with minimum eigenvalue  $> \sqrt{\lambda}$ , with  $(n+1)\epsilon < 1$ . Suppose  $b$  is a floating-point vector of length  $n$ . Suppose  $\hat{x}$  is the result of floating-point Cholesky factorization of  $A$  into  $\hat{R}^T \hat{R}$  followed by forward substitution finding  $\hat{y}$  such that  $\hat{R}^T \hat{y} = b$  and then  $\hat{R}\hat{x} = \hat{y}$ . Taking into account underflows in the factorization, but not yet considering underflows in the substitutions, we have that there is a backward error  $\delta A$  such that  $(A + \delta A)\hat{x} = b$ , bounded as follows:

$$|\delta A| \leq ((3n+1)\epsilon + n^2\epsilon^2) \frac{a_{\max} + 2(n-1)\lambda\epsilon}{1 - (n+1)\epsilon} + 4\lambda\epsilon(n+1 + a_{\max}). \quad (\text{csolve\_back\_err})$$

**Proof.** If it were not for the underflow terms  $\delta b$  and  $\delta y$ , we would have

$$\begin{aligned} b &= (\hat{R}^T + \delta R^T)\hat{y} \\ &= (\hat{R}^T + \delta R^T)(\hat{R} + \delta R)\hat{x} \\ &= (\hat{R}^T \hat{R} + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R)\hat{x} \\ &= (A + \delta A' + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R)\hat{x} \\ &= A\hat{x} + (\delta A' + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R)\hat{x}. \end{aligned} \quad (\text{xxx})$$

We will define

$$\begin{aligned} \delta A &= \delta A' + \delta R^T \hat{R} + \hat{R}^T \delta R + \delta R^T \delta R \\ &\leq |\delta A'| + |\delta R^T \hat{R}| + |\hat{R}^T \delta R| + |\delta R^T \delta R|. \end{aligned} \quad (\text{xxx})$$

From (2.3aux3) we have  $|\delta A'_{ij}| < (\min(i, j) + 2)\epsilon \|R_i\|_2 \|R_j\|_2 + 4\lambda\epsilon(n+1 + a_{\max})$ .

From csolve\_back\_err\_aux1 we have  $|(\delta R^T \hat{R})_{ij}| \leq n\epsilon(a_{\max} + 2(n-1)\lambda\epsilon)/(1 - (n+1)\epsilon)$

and similarly  $|(\hat{R}^T \delta R)_{ij}| \leq n\epsilon(a_{\max} + 2(n-1)\lambda\epsilon)/(1 - (n+1)\epsilon)$ .

From csolve\_back\_err\_aux2 we have  $|\delta R^T \delta R| \leq n^2\epsilon^2(a_{\max} + 2(n-1)\lambda\epsilon)/(1 - (n+1)\epsilon)$ .

The result follows immediately. Q.E.D.

**Still to do:** Account for underflows in the forward substitution and back substitution.

## References

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